

Thermally driven surface phase separation in intermetallic alloys

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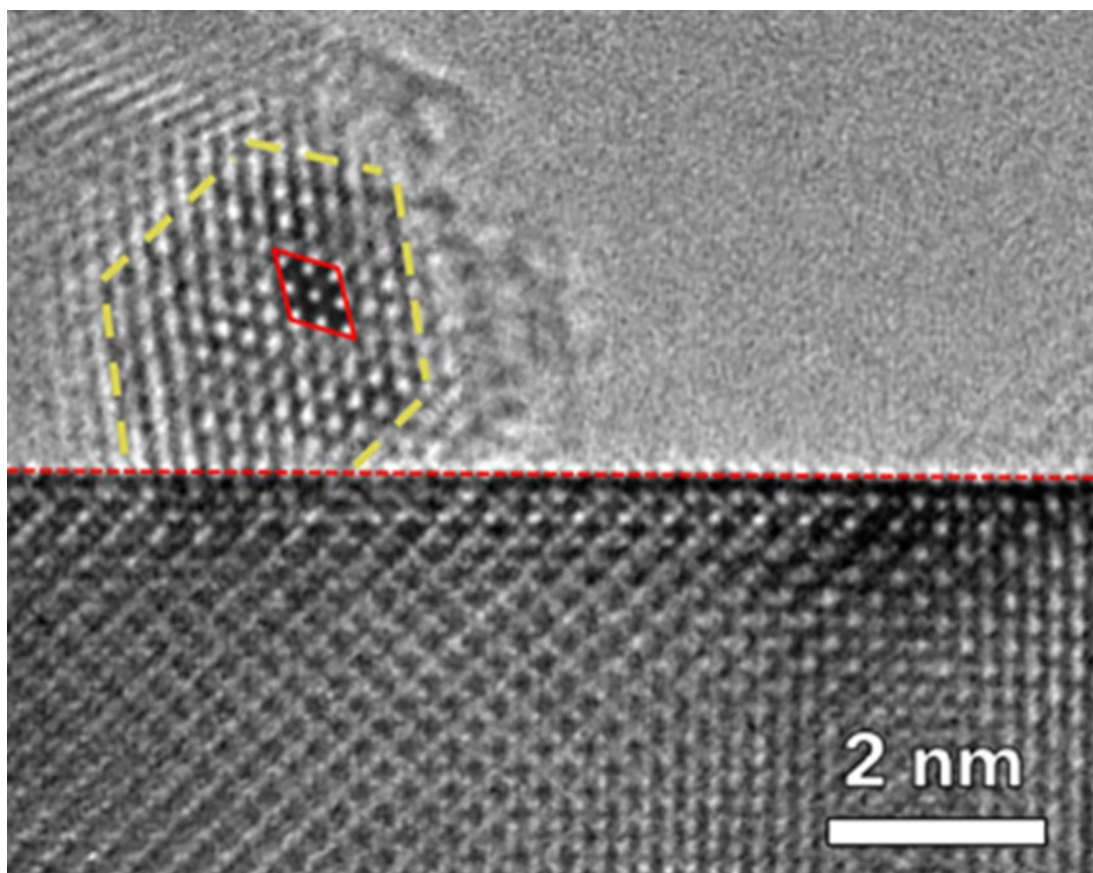
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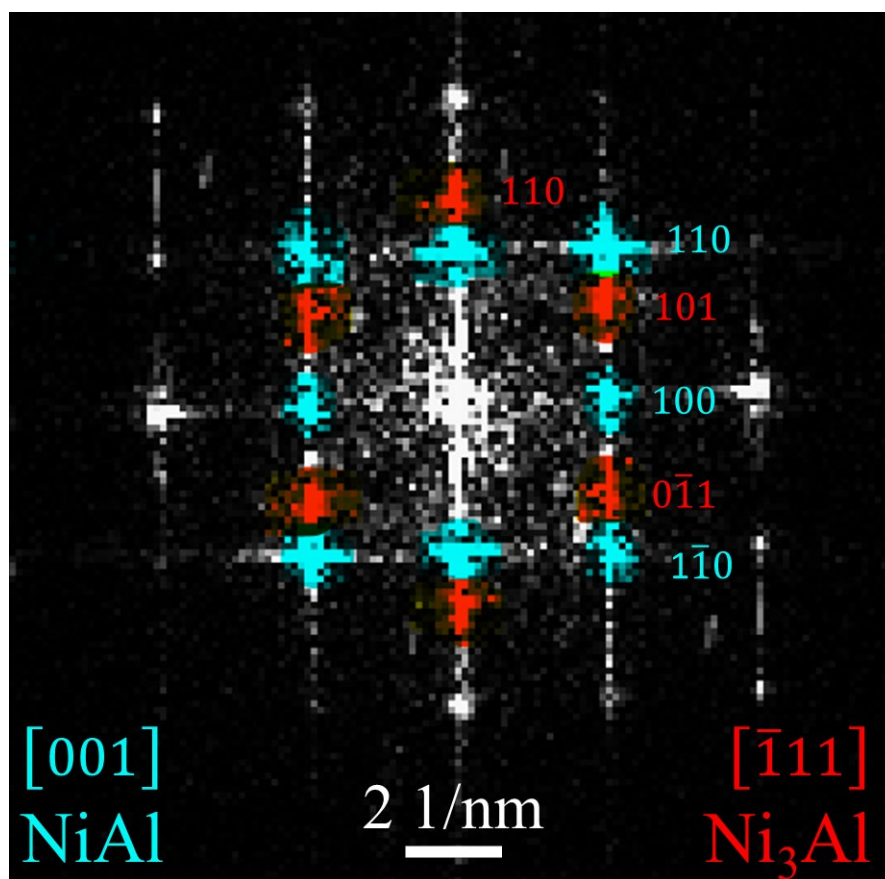
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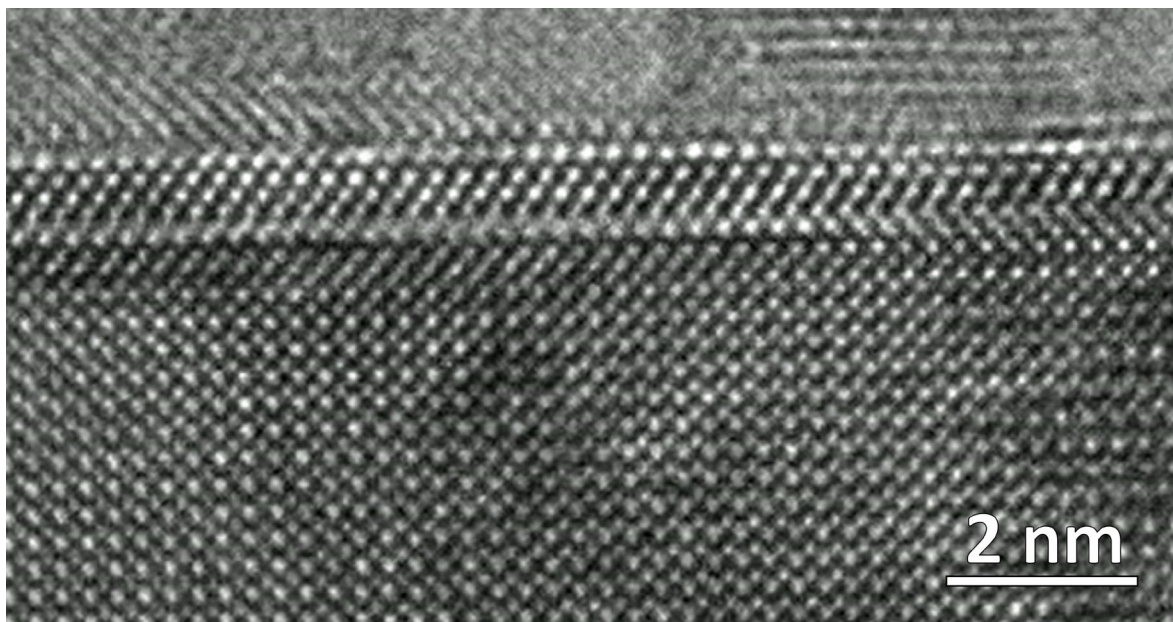
Supplementary Figures 1-6



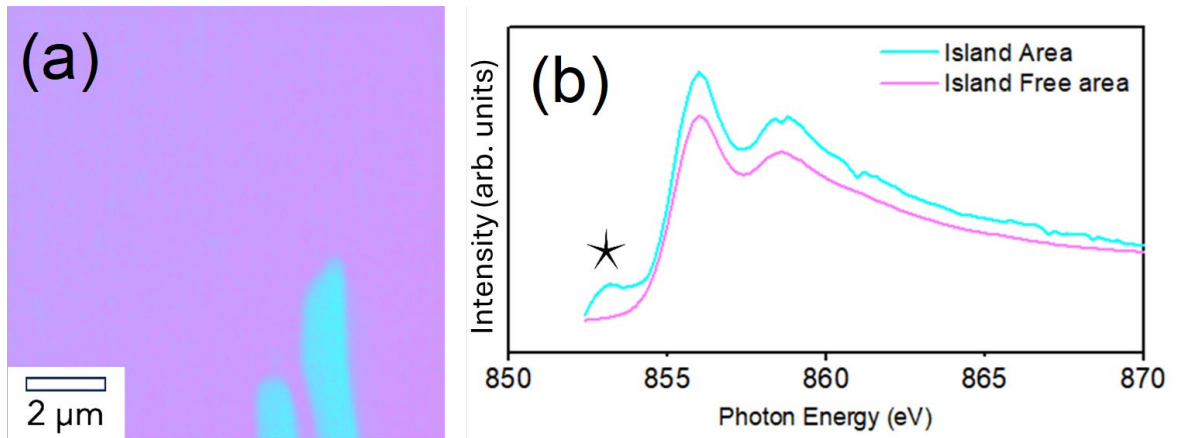
Supplementary Fig. 1. HRTEM image showing a γ' -Ni₃Al precipitate formed on the (110) facet of β -NiAl after prolonged annealing. The precipitate and interface are outlined by dashed yellow and red lines, respectively. The inset outlined in red shows an overlay of a simulated HRTEM image of γ' -Ni₃Al along the [111] zone axis.



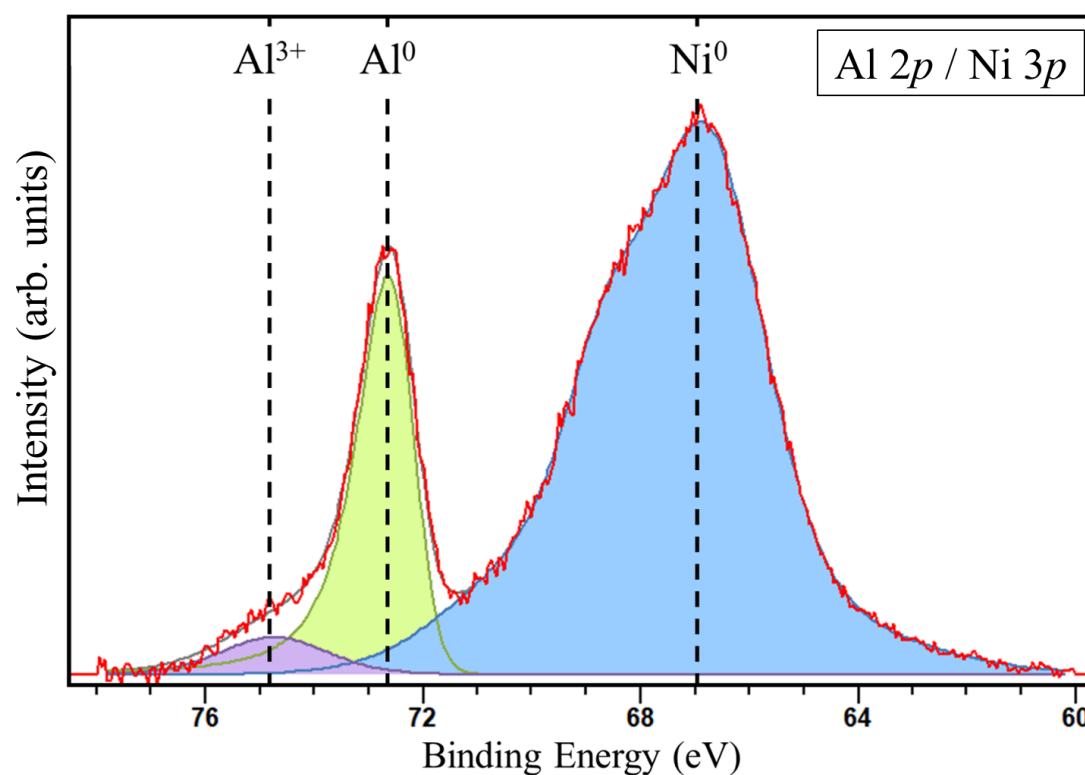
Supplementary Fig. 2. FFT analysis of the HRTEM image in Fig. 2(b). A virtual aperture selecting the NiAl(100), NiAl(110) and γ' -Ni₃Al(101) reflections are used to generate the IFFT image shown in Fig. 2(c). A second virtual aperture selecting the β -NiAl(1 $\bar{1}$ 0) and γ' -Ni₃Al(0 $\bar{1}$ 1) reflections are used to generate the IFFT image in Fig. 2(d).



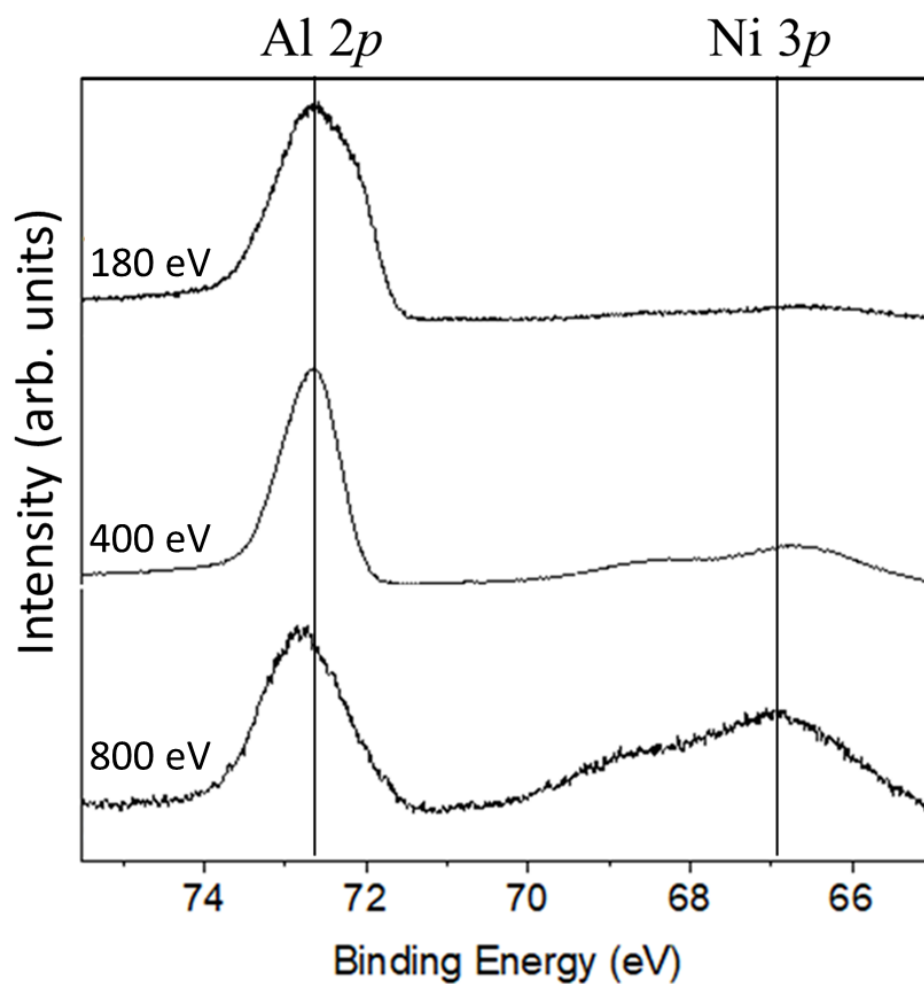
Supplementary Fig. 3. Uncolored full-field HRTEM image of a γ' -Ni₃Al precipitate overlayer formed on the (100) facet of β -NiAl after annealing at 800 °C in 0.001 Torr H₂ for ~1 h, corresponding to Fig. 2(b-d).



Supplementary Fig. 4. Secondary electron imaging and site-specific spectroscopy revealing the formation and electronic signature of a γ' -Ni₃Al island on the β -NiAl(100) surface during UHV annealing of β -NiAl(100) at 800 °C. (a) Spatially resolved image showing the formation of a γ' -Ni₃Al island (cyan) during UHV annealing of β -NiAl(100) at 800 °C. The image was acquired using secondary electrons generated by X-ray illumination (photon energy = 853.4 eV). (b) Site-specific Ni L₃-edge NEXAFS spectra collected from the γ' -Ni₃Al island region (cyan) and the island-free NiAl(100) surface (magenta) at 800 °C. The star highlights a new low-energy peak corresponding to γ' -Ni₃Al. Spectra are vertically offset for visual clarity.



Supplementary Fig. 5. XPS spectrum of the Al 2p and Ni 3p region collected from a clean β -NiAl(100) single crystal at room temperature. Peak fitting reveals a near-stoichiometric atomic composition of 49 at. % Ni and 51 at.% Al.



Supplementary Fig. 6. XPS of a clean β -NiAl(100) surface taken at 800 °C under UHV before the formation of γ' -Ni₃Al precipitates, showing the variations in the Al-to-Ni ratio in the bulk and subsurface regions as probed at photon energies of 180 eV, 400 eV, and 800 eV. These results indicate a thermodynamic preference for Al at the surface on the β -NiAl(100) matrix.